

Project Title

Loan Approval Prediction Using Machine Learning

1. Introduction

Loan Approval Prediction is a Machine Learning project used to predict whether a customer is eligible for a loan or not based on applicant details.

2. Objective

The objective of this project is to automate the loan approval process using Machine Learning algorithms.

3. Problem Statement

Banks receive many loan applications daily. Manual verification takes time. This system helps predict loan approval quickly and accurately.

4. Technologies Required

- Python
- Jupyter Notebook / VS Code
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

5. Dataset Description

The dataset contains applicant details like income, education, loan amount, credit history, property area, and loan status.

6. Methodology

1. Data Collection
2. Data Preprocessing
3. Handling Missing Values
4. Data Visualization
5. Model Training

6. Prediction
7. Accuracy Evaluation

7. Algorithm Used

Logistic Regression

8 Advantages

- Fast loan approval process
- Reduces manual work
- Improves prediction accuracy
- Easy to use

9.Conclusion

This project helps financial organizations predict loan approval efficiently using Machine Learning techniques.

10.Future Scope

Future improvements include using advanced algorithms like Random Forest and Deep Learning for better accuracy.